Assignment 1 — The Technique Ladder

(Lecture: Prompt Engineering Foundations)
Goal: Prove you can pick the *cheapest technique that works* — not the fanciest.
Task
Choose ONE real business task from your own work (or use: *"Read a customer complaint email about a billing error and produce a reply + an internal action recommendation"*).

- **Run the task through 4 techniques:** zero-shot, few-shot (2–3 examples you write yourself), chain-of-thought, and role + constraints combined. Use the exact same underlying task input each time.
- **Define success first:** before running anything, write 3 scoring criteria (e.g., accuracy, tone, actionability) with a 1–5 scale for each. Do not change them afterwards.
- **Score all 4 outputs** against your criteria in a table.
- **Break one technique:** construct an input where your winning technique fails or a "lower" technique beats a "higher" one (e.g., few-shot overfits to surface pattern; CoT produces fluent nonsense). Show the transcript.

Deliverables

- The 4 prompts + 4 raw outputs (unedited)
- Scoring rubric (written first — say so) + filled score table
- The failure case transcript + 3–5 sentences on *why* it failed
- One paragraph: which technique would you ship for this task, and what does it cost (tokens/latency) vs. the alternatives?

Grading (100 pts)

Criterion	Pts
Rubric written before running; criteria concrete, not "good/bad"	20
All 4 techniques correctly implemented (few-shot examples are real examples, CoT elicits actual reasoning, etc.)	30
Failure case is genuine and correctly diagnosed	25
Ship decision justified by scores + cost, not vibes	25

Submission

Submit a single PDF or repo. Include model name/version and temperature for every run. Raw outputs must be unedited — cherry-picking a lucky run counts against the honesty criteria.